

CHEONAN, Korea, Republic of

WMO: 471450

Lat: 36.7625N

Lon: 127.2928E

Elev: 26

StdP: 101.02

Time Zone: 9.00 (E09)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-10.7	-20.4	0.6	-7.3	-18.2	0.8	-6.9	6.7	-3.0	6.2	-2.4	1.1	270	0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.6	32.5	24.6	31.3	24.0	30.2	23.4	26.1	29.9	25.5	29.2	24.9	28.4	2.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	20.2	28.1	24.4	19.4	27.6	23.8	18.7	27.1	81.1	30.1	78.5	29.2	76.2	28.5	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	5.5	4.8	-15.7	34.5	2.8	1.2	-17.7	35.4	-19.4	36.0	-20.9	36.7	-22.9	37.5		
			-16.2	26.9	2.6	1.1	-18.1	27.7	-19.6	28.3	-21.1	28.9	-23.0	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	-2.8	0.5	5.8	11.7	18.1	22.5	25.6	25.9	20.8	14.2	7.3	-0.3
	DBStd	10.43	4.20	4.05	3.73	3.38	2.63	1.94	1.93	2.45	2.56	3.34	4.46	4.28
	HDD10.0	1257	396	266	140	22	0	0	0	0	7	106	321	
	HDD18.3	2824	654	498	387	198	36	1	0	7	133	332	579	
	CDD10.0	2176	0	1	11	73	252	376	483	494	324	137	24	0
	CDD18.3	701	0	0	0	0	29	126	224	236	81	4	0	0
	CDH23.3	5723	0	0	1	23	307	876	1869	2180	446	20	1	0
	CDH26.7	1669	0	0	0	1	44	206	564	789	66	0	0	0
Wind	WSAvg	1.8	1.7	1.9	2.3	2.3	2.0	1.7	1.6	1.6	1.5	1.5	1.8	1.7
Precipitation	PrecAvg	1267	20	29	44	76	89	148	300	278	157	53	44	29
	PrecMax	1863	58	90	113	188	206	366	644	911	379	184	154	55
	PrecMin	753	1	2	8	18	16	23	108	50	15	2	4	7
	PrecStd	274	16	22	26	39	42	93	120	182	113	46	33	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	15.7	21.3	26.2	29.9	31.2	33.2	34.8	30.2	25.5	21.1	12.7
		MCWB	5.4	11.6	13.1	14.0	17.2	19.9	24.3	24.7	22.9	18.7	15.1	10.0
	2%	DB	7.1	11.6	18.4	23.3	27.9	30.0	32.0	33.2	28.4	23.5	18.0	9.8
		MCWB	3.6	6.4	10.8	12.7	16.2	19.8	24.2	25.0	21.3	16.8	13.5	6.4
	5%	DB	5.3	9.3	15.6	21.3	26.5	28.9	30.8	32.0	27.1	21.8	16.2	8.1
		MCWB	2.5	4.9	9.1	11.6	16.1	19.5	24.1	24.8	20.3	15.6	12.1	5.5
	10%	DB	3.7	7.0	13.2	19.2	25.1	27.8	29.8	30.8	26.0	20.4	14.2	6.3
		MCWB	1.2	3.3	7.7	11.1	15.3	19.4	23.9	24.6	19.3	14.8	10.6	3.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	12.9	14.1	16.0	21.3	23.3	26.5	27.7	25.0	21.0	16.6	10.2
		MCDB	8.7	14.8	16.8	22.4	24.9	27.0	30.0	31.8	28.8	23.4	19.6	12.5
	2%	WB	4.5	7.4	12.3	14.6	18.7	22.5	25.7	26.5	23.3	19.1	14.7	7.5
		MCDB	6.6	9.8	16.6	19.9	23.6	26.4	29.5	30.2	26.2	21.8	16.9	9.2
	5%	WB	2.8	5.3	10.2	13.3	17.7	21.9	25.2	25.8	22.4	17.2	12.7	5.6
		MCDB	4.7	8.7	14.8	18.4	22.9	26.1	28.8	29.5	25.2	20.0	15.2	7.7
	10%	WB	1.4	3.8	8.1	12.2	16.8	21.0	24.8	25.2	21.3	15.7	11.3	3.9
		MCDB	3.7	6.6	12.2	17.4	22.3	25.0	28.2	28.9	24.0	18.9	14.0	6.1
Mean Daily Temperature Range	5% DB	MDBR	9.3	9.5	11.1	12.4	12.1	9.9	7.0	7.6	9.4	11.5	9.7	8.7
		MCDBR	10.3	12.8	14.9	16.6	14.7	12.0	9.0	9.6	10.6	12.6	11.9	10.1
		MCWBR	7.0	8.0	8.2	7.4	5.4	3.9	3.2	3.3	4.2	5.9	6.7	7.6
	5% WB	MCDBR	8.8	10.8	12.4	11.9	10.6	8.1	6.6	7.8	7.3	9.5	9.2	9.3
		MCWBR	6.6	7.8	7.7	6.6	5.3	3.7	2.9	3.4	3.6	5.6	6.1	7.7
Clear-Sky Solar Irradiance	taub	0.440	0.487	0.571	0.603	0.615	0.685	0.639	0.575	0.501	0.486	0.471	0.436	
	taud	1.938	1.863	1.710	1.689	1.705	1.627	1.777	1.929	2.066	1.979	1.959	1.967	
	Ebn at Noon	699	719	697	706	707	657	685	720	754	713	663	670	
	Edh at Noon	142	173	221	238	237	257	219	185	153	152	139	130	
All-Sky Solar Radiation	RadAvg	2.47	3.24	4.26	4.95	5.56	5.11	4.15	4.33	4.04	3.57	2.48	2.19	
	RadStd	0.14	0.27	0.43	0.39	0.50	0.46	0.59	0.62	0.45	0.28	0.28	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (53 neighbors)	N/A	N/A	-0.81	N/A	N/A	N/A	N/A	N/A	+87	+47

Nomenclature: See separate page